

Optimizing MULTI-UAV Cooperative Task Assignment for Fire Extinguishing Missions

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Abstract—Forest fires represent a growing global threat, with millions of hectares lost annually due to their rapid spread and devastating impact on ecosystems, biodiversity, and human settlements. The increasing frequency and intensity of such disasters motivate the need for intelligent and automated solutions capable of responding swiftly and efficiently. In this context, cooperative task assignment for unmanned aerial vehicles (UAVs) emerges as a promising approach to enhance aerial firefighting and surveillance operations. However, this remains a challenging problem due to its combinatorial nature, multiple conflicting objectives, and operational constraints. This work formulates the UAV–task assignment problem as a multi-objective optimization model that maximizes task value while minimizing UAV losses and the number of deployed UAVs. To solve this problem, four meta-heuristic algorithms are investigated: Simulated Annealing (SA), Genetic Algorithm (GA), Discrete Grey Wolf Optimizer (DGWO), and Binary Firefly Algorithm (BFF). Each method is adapted to operate directly on binary assignment matrices with appropriate constraint-handling mechanisms. Simulation results demonstrate that the Binary Firefly Algorithm (BFF) achieves the best overall performance in terms of solution quality and consistency, while Simulated Annealing (SA) exhibits the fastest convergence speed among the compared methods. Overall, the evaluated algorithms effectively produce high-quality feasible task allocations, though runtime trade-offs must be considered for time-critical wildfire deployment scenarios.

Index Terms—Unmanned Aerial Vehicles, Task Assignment, Meta-heuristic Optimization, Simulated Annealing, Genetic Algorithm, Discrete Grey Wolf Optimizer, Binary Firefly Algorithm

I. INTRODUCTION

The increasing severity of climate-driven wildfires has intensified the need for automated response strategies. Unmanned aerial vehicles (UAVs) provide a versatile solution for real-time monitoring in hazardous environments where human intervention is restricted. Consequently, optimizing cooperative task assignment within UAV swarms is essential to enhancing operational efficiency and ensuring fleet survivability [1], [2].

The UAV–task assignment problem involves distributing multiple UAVs across fire zones while balancing mission coverage against constraints like fuel capacity and flight range. As the scale of the problem increases, the combinatorial complexity makes exact solutions computationally intensive. To address this, meta-heuristic optimization is used to explore the large solution space efficiently. This study evaluates four

algorithms—Simulated Annealing (SA), Genetic Algorithm (GA), Discrete Grey Wolf Optimizer (DGWO), and Binary Firefly Algorithm (BFF)—all adapted for binary assignment matrices.

SA is a probabilistic search inspired by the physical cooling of materials, exploring neighboring solutions and probabilistically accepting worse solutions early to escape local optima [3]. GA maintains a population of candidate solutions, applying selection, crossover, and mutation to evolve better assignments over generations. Due to their simplicity, robustness, and strong global search capabilities, both SA and GA are widely employed as benchmark algorithms in comparative optimization studies [4], [5].

DGWO is inspired by the social hierarchy and cooperative hunting behavior of grey wolf packs. Wolves are ranked as alpha, beta, delta, and omega, representing the best, second-best, third-best, and remaining solutions. The search process is guided by the top three wolves, while the rest of the pack update their positions relative to them. In this study, DGWO is fully adapted for the binary assignment domain, ensuring that all generated solutions are feasible while retaining the balance between exploration and exploitation inherent in the algorithm. Comparative studies have consistently demonstrated that DGWO achieves superior objective values compared to other metaheuristics, which justifies its selection as the optimization algorithm for this task assignment problem [6].

This work implements the Binary Firefly Algorithm (BFF), which mimics the attraction-based movement of fireflies to explore and exploit promising regions of the assignment matrix. Alongside SA, GA, and DGWO, the BFF is evaluated based on convergence behavior and computational efficiency. The resulting comparison highlights the strengths of each meta-heuristic, offering a framework for more effective UAV-based wildfire management.

II. METHODOLOGY

A. Problem Formulation

This paper looks at making a plan to assign unmanned aerial vehicles (UAVs) to fire zones for wildfire suppression operations. In real situations, the goal is to extinguish as many fire zones as possible, but fire zones have different levels of severity and priority. Addressing a high-priority fire zone can

strongly affect the overall outcome of the firefighting mission, yet those zones are usually more intense, harder to suppress, and riskier for the deployed UAVs. The main challenge is to match UAVs to fire zones so that zone priority, fire intensity, expected UAV losses, and the number of UAVs deployed are all balanced. The aim is to create an allocation plan that maximizes firefighting effectiveness while keeping losses and resource use low [7].

Let $I_n = \{1, 2, \dots, n\}$, $\bar{I}_n = \{0\} \cup I_n$, and $\mathbb{N}_+ = \{1, 2, 3, \dots\}$, where $n \in \mathbb{N}_+$. At the environment, a group of unmanned aerial vehicles (UAVs) is available to carry out coordinated firefighting on a number of known fire zones. Assume that N_U ($N_U \in \mathbb{N}_+$) UAVs are selected from a total of N_{total} ($N_{\text{total}} \in \mathbb{N}_+$) UAVs in our available fleet to jointly suppress N_T ($N_T \in \mathbb{N}_+$) fire zones located in the wildfire area.

Let $U = \{U_1, U_2, \dots, U_{N_U}\}$ be the set of UAVs selected for the mission, where U_i ($i \in I_{N_U}$) represents the i -th UAV. These UAVs are chosen from the total UAV pool

$$U_{\text{total}} = \{U_1, U_2, \dots, U_{N_{\text{total}}}\}, \quad (1)$$

where $U \subseteq U_{\text{total}}$ and $|U| = N_U$.

The task set is denoted by

$$T = \{T_1, T_2, \dots, T_{N_T}\}, \quad (2)$$

where T_j ($j \in I_{N_T}$) represents the j -th task. When a UAV U_i suppresses fire zones T_j , there is a possibility that the UAV may be destroyed. Different UAVs have different probabilities of being destroyed when suppressing different fire zones. Let P_{ij} and K_{ij} respectively denote the probabilities that UAV U_i and fire zone T_j are destroyed when U_i suppresses T_j . Furthermore, let V_{T_j} ($j \in I_{N_T}$) represent the value (or importance) of fire zone T_j , and let W_{U_i} ($i \in I_{N_U}$) represent the value (or cost) of UAV U_i . [8]

Thus, the decision problem is to select the subset $U \subseteq U_{\text{total}}$ and assign each UAV $U_i \in U$ to a fire zone $T_j \in T$ in order to maximize overall mission effectiveness—such as the expected total value of extinguish fire zones—while considering UAV survivability, potential losses, and available resources. Let C_i represent the maintenance cost of each participating UAV.

1) Objective Functions

The first objective is to maximize the total value of the fire zones extinguished by the selected UAVs:

$$\max f_1(U, x) = \sum_{i=1}^{N_U} \sum_{j=1}^{N_T} K_{ij} V_{T_j} x_{ij} \quad (3)$$

The second objective is to minimize the total expected cost of UAVs lost during the mission:

$$\min f_2(U, x) = \sum_{i=1}^{N_U} \sum_{j=1}^{N_T} P_{ij} W_{U_i} x_{ij} \quad (4)$$

The third objective is to minimize the number of UAVs that participate in suppression:

$$\min f_3(U) = \sum_{i=1}^{N_U} C_i \quad (5)$$

The three objectives are combined into a single weighted scalar objective for optimization:

$$F(x) = w_1 f_1(U, x) - w_2 f_2(U, x) - w_3 f_3(U, x), \quad (6)$$

where w_1 , w_2 , and w_3 are non-negative weights that reflect the relative importance of each objective. Two weight configurations are evaluated: $(w_1, w_2, w_3) = (50, 1000, 2000)$ for individual algorithm convergence analysis, and $(w_1, w_2, w_3) = (5, 1000, 20000)$ for the comparative evaluation, where the higher penalty on f_3 more strongly discourages unnecessary UAV deployment.

2) Decision Variables

From the objective functions above, the decision variables are:

The binary decision variable x_{ij} is defined as:

$$x_{ij} = \begin{cases} 1, & \text{if UAV } U_i \text{ is assigned to fire zone } T_j, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

The set of participating UAVs is:

$$U_{\text{sel}} = \{U_i \in U_{\text{total}} \mid \exists j : x_{ij} = 1\}. \quad (8)$$

3) Constraints

The first constraint is number of tasks assigned to each UAV cannot exceed its own ammunition capacity:

$$\sum_{j=1}^{N_T} x_{ij} \leq n_i, \quad \forall i \in I_{N_U}, \quad (9)$$

The second constraint is number of tasks for each fire zone is limited:

$$\sum_{i=1}^{N_U} x_{ij} \leq m_j, \quad \forall j \in I_{N_T}, \quad (10)$$

The third constraint ensures that the number of UAVs participating in the mission is compatible with the number of fire zones. From the algorithmic perspective, there is a desired number of UAVs, denoted as N_{wanted} , which represents the optimal or preferred number of UAVs calculated by the algorithm:

$$\frac{N_T}{n} \leq N_U \leq N_{\text{total}}, \quad (11)$$

The fourth constraint ensures that each fire zone is assigned to at least one UAV. In other words, every task in the wildfire area must be suppressed by one or more UAVs to guarantee mission completeness and prevent any fire zone from being left unengaged:

$$\sum_{i=1}^{N_U} x_{ij} \geq 1, \quad \forall j \in I_{N_T} \quad (12)$$

After formulating the problem, solution visualization is necessary to verify that the results produced by the algorithms are correct, as illustrate in Fig.1

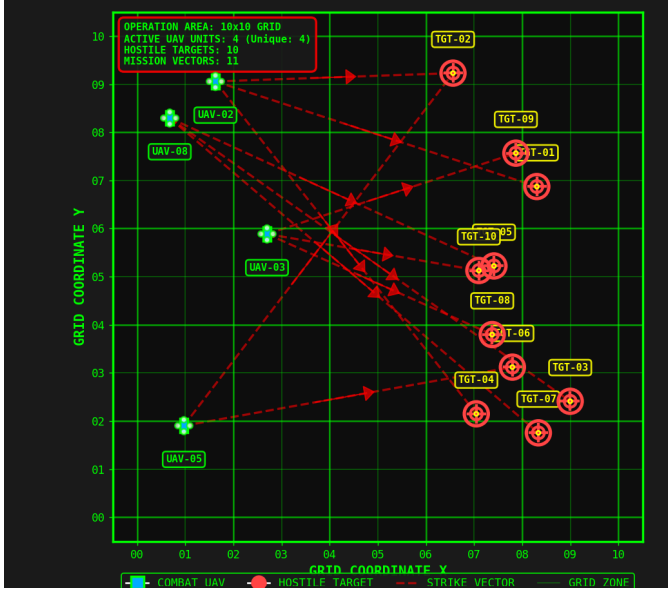


Fig. 1. Tactical map and UAV-fire zone assignment visualization. Fire zones are marked by colored circles scaled to their priority value V_{T_j} ; UAVs are shown as labeled markers at their base positions; colored lines indicate assignment of each UAV to its designated fire zone(s).

B. Discrete Grey Wolf Optimizer (DGWO)

Algorithm 1 Discrete Grey Wolf Optimizer (DGWO) for UAV-task Assignment

```

1: Initialize wolf pack  $\mathcal{W} \leftarrow \text{randomgeneration}()$ 
2: for each wolf  $(X, U) \in \mathcal{W}$  do
3:    $X \leftarrow \text{repair\_solution}(X)$ 
4:   Evaluate fitness  $f(X)$ 
5: end for
6: for  $t = 1$  to  $T_{\max}$  do
7:   Identify leaders  $\alpha, \beta, \delta$ 
8:   for each wolf  $(X, U) \in \mathcal{W}$  do
9:     Compute distances  $D_\alpha, D_\beta, D_\delta$  and normalize  $P_1, P_2, P_3$ 
10:     $P \leftarrow (P_1 + P_2 + P_3)/3$ 
11:    Compute coefficients  $A_\alpha, A_\beta, A_\delta$ 
12:    Generate candidates  $X'_\ell \leftarrow \text{DiscreteCrossover}(X, X_\ell, A_\ell), \ell \in \{\alpha, \beta, \delta\}$ 
13:    Combine candidates via majority voting to obtain  $X'$ 
14:    Apply discrete operator based on  $P$ , repair, and evaluate  $f(X')$ 
15:   end for
16: end for
17: return best wolf  $\alpha$ 

```

The classical Grey Wolf Optimizer (GWO) updates search agents in a continuous domain using three leaders (α, β, δ) to guide the population. However, the UAV-task assignment problem is inherently *binary*:

$$x_{ij} \in \{0, 1\} \quad (13)$$

and therefore the continuous GWO update rule cannot be used directly. To address this limitation, we employ the discrete

mapping mechanism proposed in the Discrete Improved Grey Wolf Optimizer (DGWO) [9].

C. Simulated Annealing (SA)

Algorithm 2 Simulated Annealing for UAV-task Assignment

```

1: Input:  $T_0, T_f, I_{\max}$ 
2:  $x_{\text{cur}} \leftarrow \text{randomgeneration}()$ 
3:  $f_{\text{cur}} \leftarrow \text{the\_total\_objective}(x_{\text{cur}})$ 
4:  $x_{\text{best}} \leftarrow x_{\text{cur}}, f_{\text{best}} \leftarrow f_{\text{cur}}$ 
5: for  $k = 1$  to  $I_{\max}$  do
6:    $x_{\text{new}} \leftarrow \text{randomchange}(x_{\text{cur}})$ 
7:   if Feasability( $x_{\text{new}}$ ) then
8:      $f_{\text{new}} \leftarrow \text{the\_total\_objective}(x_{\text{new}})$ 
9:      $\Delta E \leftarrow f_{\text{new}} - f_{\text{cur}}$ 
10:    if  $\Delta E > 0$  or  $\text{rand}() < e^{\Delta E/T}$  then
11:       $x_{\text{cur}} \leftarrow x_{\text{new}}, f_{\text{cur}} \leftarrow f_{\text{new}}$ 
12:      if  $f_{\text{cur}} > f_{\text{best}}$  then
13:         $x_{\text{best}} \leftarrow x_{\text{cur}}, f_{\text{best}} \leftarrow f_{\text{cur}}$ 
14:      end if
15:    end if
16:  end if
17:   $T \leftarrow \max(T_f, T_0 \cdot 0.95^k)$ 
18: end for
19: Return:  $x_{\text{best}}, f_{\text{best}}$ 

```

D. Genetic Algorithm (GA)

Algorithm 3 Genetic Algorithm for UAV-task Assignment

```

1: Input: Population size  $N$ , crossover rate  $p_c$ , mutation rate  $p_m$ , max generations  $G_{\max}$ 
2: Initialize population  $\mathcal{P} \leftarrow \text{randomgeneration}()$ 
3: for each  $x \in \mathcal{P}$  do
4:    $x \leftarrow \text{repair\_solution}(x)$ 
5:    $f(x) \leftarrow \text{the\_total\_objective}(x)$ 
6: end for
7:  $\mathcal{P}_{\text{best}} \leftarrow \mathcal{P}$ 
8: for  $g = 1$  to  $G_{\max}$  do
9:    $\mathcal{M} \leftarrow \text{selection}(\mathcal{P})$ 
10:   $\mathcal{C} \leftarrow \text{crossover}(\mathcal{M}, p_c)$ 
11:   $\mathcal{C} \leftarrow \text{mutation}(\mathcal{C}, p_m)$ 
12:  for  $x \in \mathcal{C}$  do
13:     $x \leftarrow \text{repair\_solution}(x)$ 
14:     $f(x) \leftarrow \text{the\_total\_objective}(x)$ 
15:  end for
16:   $\mathcal{P} \leftarrow \text{elitism\_replacement}(\mathcal{P}, \mathcal{C})$ 
17:  Update  $\mathcal{P}_{\text{best}}$  if improved
18: end for
19: Return:  $x_{\text{best}}, f_{\text{best}}$ 

```

E. Binary Firefly Algorithm

In Algorithm 4, $\text{hammering_beta}(x_i, x_j)$ computes the Hamming distance d_H between two fireflies:

$$d_H(x_i, x_j) = \sum_{k=1}^{N_U \times N_T} |x_i^k - x_j^k|, \quad (14)$$

and the corresponding attractiveness β :

$$\beta = \beta_0 e^{-\gamma d_H(x_i, x_j)^2}, \quad (15)$$

where β_0 is the initial attractiveness and γ is the light absorption coefficient. The function `firefly_move`(x_i, x_j, β) moves x_i toward x_j by probabilistically flipping bits where the two solutions differ, with flip probability proportional to β :

$$x_i^k \leftarrow \begin{cases} 1 - x_i^k, & \text{if } \text{rand}() < \beta \text{ and } x_i^k \neq x_j^k, \\ x_i^k, & \text{otherwise,} \end{cases} \quad (16)$$

followed by a random perturbation step scaled by α :

$$x_i^k \leftarrow \begin{cases} 1 - x_i^k, & \text{if } \text{rand}() < \alpha, \\ x_i^k, & \text{otherwise.} \end{cases} \quad (17)$$

Algorithm 4 Binary Firefly Algorithm for UAV-task Assignment [10]

```

1: Input:  $N, G_{\max}, \beta_0, \gamma, \alpha, \delta$ 
2: Initialize population  $\mathcal{P} \leftarrow \text{randomgeneration}()$ 
3: for  $x \in \mathcal{P}$  do
4:    $x \leftarrow \text{repair\_solution}(x), f(x) \leftarrow \text{the\_total\_objective}(x)$ 
5: end for
6:  $x_{\text{best}} \leftarrow \text{best in } \mathcal{P}$ 
7: for  $g = 1$  to  $G_{\max}$  do
8:   for  $x_i \in \mathcal{P}$  do
9:     for  $x_j \in \mathcal{P}$  do
10:      if  $f(x_j) > f(x_i)$  then
11:        Compute Hamming distance and
         $\beta \leftarrow \text{hammering\_beta}(x_i, x_j)$ 
12:         $x_i \leftarrow \text{firefly\_move}(x_i, x_j, \beta)$ , repair and
        enforce uniqueness
13:        Update  $f(x_i)$ 
14:      end if
15:    end for
16:  end for
17:  Apply probabilistic mutation to  $\mathcal{P}$ 
18:  Update  $x_{\text{best}}$  if improved
19:   $\alpha \leftarrow \delta \cdot \alpha$ 
20: end for
21: Return:  $x_{\text{best}}, f_{\text{best}}$ 

```

III. RESULTS

A. Simulated Annealing (SA)

Fig.2 illustrate the convergence behavior of the Simulated Annealing (SA) algorithm for the UAV-Task Assignment problem. The first plot (Current Objective vs. Iteration) shows the typical fluctuating pattern of SA, where the objective value initially varies significantly due to the probabilistic acceptance of worse solutions, promoting exploration. As iterations progress, the fluctuations reduce and the solution quality stabilizes at a higher objective value. The second plot (Best Objective vs. Iteration) demonstrates a monotonic improvement, confirming that the best-found solution is consistently updated whenever a superior solution is discovered,

ultimately converging to a stable optimum. The third plot depicts the temperature schedule, which decreases exponentially over iterations, gradually shifting the algorithm from exploration to exploitation. Finally, the fourth plot shows the evolution of the number of active UAVs (or assignment changes), where higher variability appears in early iterations and stabilizes later, indicating convergence toward a consistent assignment structure. Overall, the graphs confirm the effective balance between exploration and exploitation achieved by the SA algorithm.

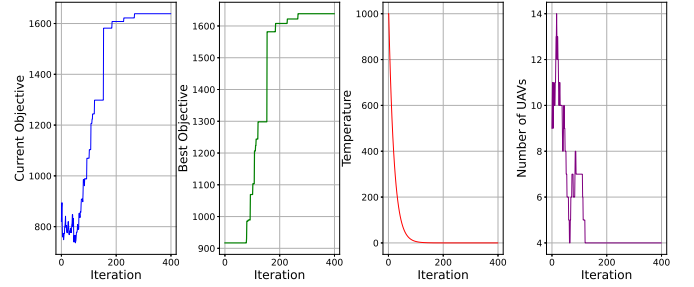


Fig. 2. SA convergence for UAV-task assignment with weights $(w_1, w_2, w_3) = (50, 1000, 2000)$ over 400 iterations. From left to right: current objective $F(x)$ vs. iteration; best objective vs. iteration; temperature T vs. iteration; number of active UAVs vs. iteration.

B. Genetic Algorithm (GA)

Fig.3 illustrates the convergence behavior of the Genetic Algorithm (GA) over generations. The average objective value rises sharply in the early generations and quickly plateaus at a high level, indicating that GA converges rapidly toward high-quality firefighting task allocation solutions with minimal improvement in later stages. The generation number increases linearly as expected, confirming the steady progression of the evolutionary process. The number of UAVs deployed drops quickly to a minimal value of 4 UAVs within the early generations and remains stable throughout the rest of the run, reflecting the algorithm's ability to efficiently minimize resource usage while still maintaining effective fire zone coverage. The best objective value similarly reaches a high level early in the run and remains stable, demonstrating that GA is capable of identifying near-optimal solutions within the first fraction of its total generations and sustaining that quality until termination. Overall, the GA exhibits fast and stable convergence with good solution quality and aggressive resource minimization, reducing the deployed fleet to its lowest feasible level very early in the optimization process.

C. Discrete Grey Wolf Optimizer (DGWO)

Fig.4 illustrates the convergence behavior of the Discrete Grey Wolf Optimizer (DGWO) across four metrics over iterations. The average fitness value rises sharply in the very early iterations and then stabilizes around 1600, though with noticeable fluctuations throughout the run, suggesting that DGWO continues to explore the search space even after convergence,

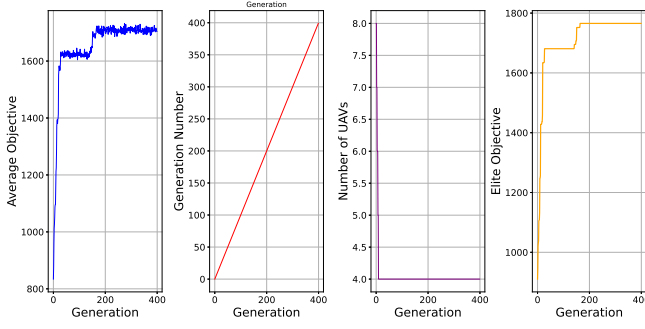


Fig. 3. GA convergence for UAV-task assignment with weights $(w_1, w_2, w_3) = (50, 1000, 2000)$, population size 100, over 400 generations. From left to right: average objective $F(x)$ vs. generation; generation index; number of deployed UAVs; best objective vs. generation.

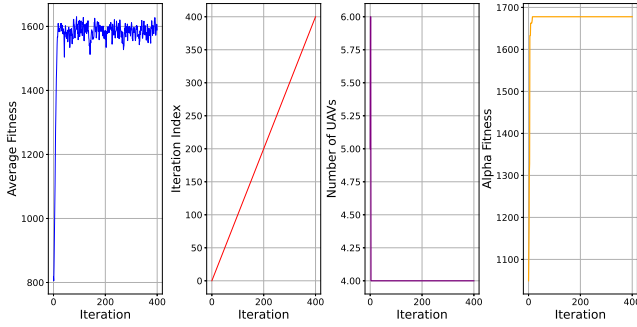


Fig. 4. DGWO convergence for UAV-task assignment with weights $(w_1, w_2, w_3) = (50, 1000, 2000)$, pack size 100, over 400 iterations. From left to right: average fitness vs. iteration; iteration index; number of deployed UAVs; alpha fitness (best solution) vs. iteration.

which reflects its balance between exploitation led by the alpha wolf and exploration by the rest of the pack. The iteration index increases linearly as expected, confirming the steady progression of the optimization process. The number of UAVs deployed drops immediately to 4 UAVs within the first few iterations and remains fixed at that value for the entire run, demonstrating that DGWO is highly efficient in minimizing resource usage from a very early stage. The alpha fitness, representing the best solution found by the leading wolf, converges rapidly to a high value of approximately 1700 within the first few iterations and remains perfectly stable thereafter, indicating that DGWO identifies its best solution very early and preserves it consistently throughout the optimization. Overall, DGWO demonstrates fast convergence in terms of best solution quality and resource minimization, while maintaining a degree of diversity in its average fitness due to the exploratory nature of the wolf pack mechanism.

D. Binary Firefly Algorithm Results

Fig 5 illustrates the convergence behavior of the Binary Firefly Algorithm (BFF) across four metrics over iterations. Unlike the other algorithms, the average objective value of BFF shows a slow and gradual improvement throughout the entire run, starting from around 800 and steadily climbing

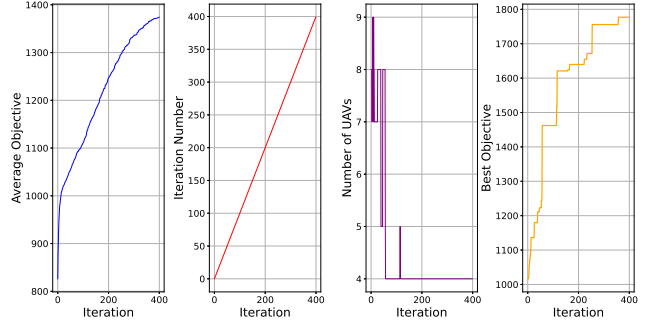


Fig. 5. BFF convergence for UAV-task assignment with weights $(w_1, w_2, w_3) = (50, 1000, 2000)$, population size 100, over 400 iterations. From left to right: average objective $F(x)$ vs. iteration; iteration index; number of deployed UAVs; best objective vs. iteration.

toward 1400 without fully plateauing, indicating that BFF continues to explore and refine solutions across all iterations rather than converging quickly. The iteration index increases linearly as expected, confirming the steady progression of the optimization process. The number of UAVs deployed exhibits high variability in the early iterations, fluctuating between 4 and 9 UAVs, before eventually settling at 4 UAVs after approximately 150 iterations, reflecting the exploratory nature of BFF in the early stages as fireflies move toward brighter companions and the swarm gradually organizes itself into a more stable allocation. The best objective value follows a staircase-like increasing pattern, improving in discrete jumps throughout the run and continuing to find better solutions even in later iterations, which highlights BFF's strong and sustained exploration capability, enabling it to keep discovering improved solutions well into the optimization process.

E. Genetic Algorithm, Simulated Annealing, Discrete Grey Wolf, and Binary Firefly Comparison

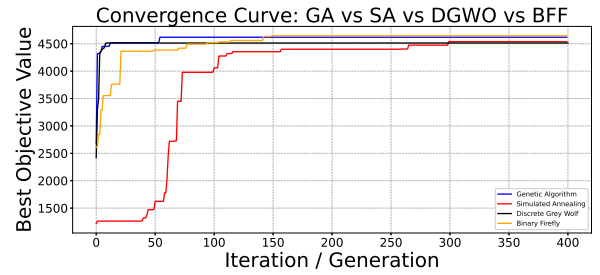


Fig. 6. Convergence comparison between the Genetic Algorithm (GA), Simulated Annealing (SA), Discrete Grey Wolf Optimizer (DGWO), and Binary Firefly Algorithm (BFF) for the UAV-task assignment problem using objective weights $(5, 1000, 20000)$ over 400 iterations.

Figure 6 compares the convergence behavior of four meta-heuristic algorithms for the UAV-task assignment problem. All methods show progressive improvement over iterations, but differ in final solution quality, convergence stability, and computational cost.

TABLE I
PERFORMANCE COMPARISON OF GA, SA, DGWO, AND BFF OVER 10
INDEPENDENT RUNS

Metric	GA	SA	DGWO	BFF
Best Objective Value	4668.59	4641.54	4608.95	4697.87
Mean Objective Value	4561.60	4521.09	4523.58	4663.79
Standard Deviation	61.20	81.66	58.23	19.52
Run Time (s)	10.20	1.74	10.53	265.31

TABLE II
BEST ASSIGNMENT MATRICES PER ALGORITHM

Algo.	UAV	Assignment Vector
GA	UAV(5)	[0,0,0,0,0,0,1,0,0]
	UAV(6)	[0,0,1,0,0,1,0,0,0,1]
	UAV(7)	[1,1,0,1,0,0,0,0,0,0]
	UAV(11)	[0,0,0,0,1,0,1,0,1,0]
SA	UAV(5)	[0,0,0,0,0,0,1,0,0,0]
	UAV(6)	[1,0,1,0,0,0,0,1,0,0]
	UAV(7)	[0,1,0,1,0,1,0,0,0,0]
	UAV(11)	[0,0,0,0,1,0,0,0,1,1]
DGWO	UAV(5)	[0,0,1,0,0,0,0,0,0,0]
	UAV(6)	[1,0,0,0,0,0,1,0,1,0]
	UAV(7)	[0,1,0,0,0,1,0,0,0,1]
	UAV(14)	[0,0,0,1,1,0,0,1,0,0]
BFF	UAV(5)	[0,0,0,0,1,0,0,0,0,0]
	UAV(6)	[1,0,1,0,0,0,1,0,0,0]
	UAV(7)	[0,1,0,0,0,1,0,0,0,1]
	UAV(11)	[0,0,0,1,0,0,0,1,1,0]

The Binary Firefly Algorithm (BFF) achieves the highest solution quality, reaching a best objective value of **4697.87** with the lowest variability (standard deviation **19.52**), indicating highly consistent convergence. However, this performance comes at a substantial computational cost, requiring **265.31 seconds** per run. The Genetic Algorithm (GA) provides a strong trade-off between solution quality and efficiency, attaining a best value of **4668.59** with an execution time of **10.20 seconds**.

Simulated Annealing (SA) is the fastest method, completing each run in only **1.74 seconds** while achieving a competitive best value of **4641.54**. The Discrete Grey Wolf Optimizer (DGWO) delivers stable and competitive performance with a best value of **4608.95** and an execution time comparable to GA (**10.53 seconds**). While BFF achieves the highest solution quality, its runtime of **265.31 s** is approximately $150\times$ slower than SA and $26\times$ slower than GA, raising practical concerns for real-time wildfire response where rapid deployment decisions are critical. For time-critical scenarios, SA or GA are more operationally viable.

IV. CONCLUSION

This paper proposed a binary meta-heuristic framework for multi-UAV cooperative task assignment in wildfire suppression, formulated as a weighted three-objective optimization problem balancing fire zone coverage, UAV survivability, and deployment efficiency. Four algorithms — SA, GA, DGWO, and BFF — were each adapted to operate

natively on binary assignment matrices through three key contributions: a majority-voting position update for DGWO, a Hamming-distance attraction mechanism for BFF, and a unified constraint-repair procedure applied consistently across all methods.

Experimental results over 10 independent runs demonstrated clear performance differentiation among the four algorithms. BFF achieved the highest solution quality with the lowest variability, confirming its strong and consistent exploration capability. GA offered the best balance between solution quality and computational efficiency, making it the most operationally viable choice for time-critical wildfire deployment scenarios. SA delivered the fastest convergence with acceptable solution quality, while DGWO produced stable and competitive results through its leader-guided discrete search mechanism.

Future work will target three directions: reducing BFF's computational overhead through parallel or GPU-accelerated implementations, developing hybrid algorithms that combine the convergence speed of SA with the exploration strength of BFF, and extending the framework to dynamic re-assignment scenarios under evolving fire conditions with heterogeneous UAV fleets.

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